

Final Report - A Pneu Robot

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I. PROJECT IDEA SYNOPSIS

Soft robotic technologies offer new avenues to approaching the construction of robotic platforms. While the applications vary widely, two common applications are terrain-compliant locomotion and robust grasping. In this report, the approach is discussed to combine these two frontiers into a single comprehensive robot. In addition to the two, durability is prioritized as a key aspect. The concepts were derived from soft actuation and soft sensing to accomplish these objectives.

The robot (in brief) was composed of eight PneuNet actuators. Four were to enable movement, three to act as a ‘grabber’, and one to connect the two components. A custom touch sensor was designed to fit within the palm of the grabber for grasp detection. Fittingly, the robot is named ‘The Pneu Robot’.

The Pneu Robot created in this project exploits many of the unique advantages of soft robotic technologies. Because of its material composition of solely silicone rubber, the robot has structural flexibility allowing it to bend and deform in response to its environment. This is useful when it comes to moving on different surfaces, such as granular surfaces. This soft robot has almost no exposed hardware or electronics, making it safer for human interaction and reducing the danger for damage or harm. This makes the robot more ideal for human-robot collaboration than a rigid robot.

The entirely silicon-based actuators take advantage of the durability such materials are known for. They are capable of stretching, bending, and applying force when inflated. With the actuators’ infinite degree of freedom, the robot is highly compliant with terrain and objects it grasps. This allows the robot to be inherently safe, tolerant, and adaptive to unknown and unpredictable environments.

II. RELEVANT WORKS

Similar robots have been constructed in lab environments fairly regularly. Our base was highly reminiscent of Harvard University’s multigait soft robot. It took inspiration from animals with no stiff components (octopuses, starfish, etc). Their robot moved in a similar fashion, inflating and deflating five separate actuators to ‘walk’ [1]. Our robot was built as a reduced version of this by only implementing four actuators as the legs and a

more rigid silicon region in the center as opposed to a fifth actuator.

The creation of the actuators was also inspired by our work in lab experiment three [3]. After fabricating our own Pneu-Nets from scratch, it became apparent that these were extremely effective and simplistic mechanisms. They proved to be key to our robot’s design, as is discussed in Section IV. The grabber and connective piece between it and the body were created with the capabilities of these actuators in mind.

Additionally, a sensor was fabricated to detect whether an object is touching the palm of the gripper. The sensor was inspired by the capacitor sensor seen during the lab experiments, however it works using resistance instead of capacitance. The construction, functionality, and characteristics of the sensor is further detailed in Section IV.

III. INTELLECTUAL MERIT

This project is inspired by the earthworm robot developed by the Italian Institute of Technology (IIT) [2], which achieves crawling locomotion through hydraulic pressure using peristaltic waves. Earthworms are naturally adapted to navigating soil, a challenge for hard robots, especially in lunar or Martian regolith, where environmental factors can hinder traditional robotic designs. Our project explores the potential of soft robotics for planetary exploration, aiming to address these challenges and reduce costs compared to hard robots. For this to succeed, an effective closed-loop pressurization system must be developed and tested on regolith. The primary objective of this iteration was to fabricate a crawling robot as a foundation for future planetary exploration adaptations.

IV. DESIGN AND MANUFACTURING

Due to the balance of flexibility, controllability, and strength of PneuNet actuators, we chose to comprise our robot primarily of these. Designing with PneuNet actuators allowed for the robot to have a worm-like motion providing a semi-quick forward motion compared to other soft robots. At a higher level, we split the problem into two main parts: locomotion and grasping.

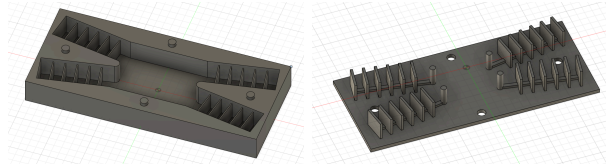


Figure 1: CAD of the body's mold

The base of the robot was modeled in Fusion 360 where molds to create the base were made, shown in Figure 1. These molds were inspired by the molds used in lab 3. The bottom mold has slits along the legs to create incisions giving each leg 'elbows' allowing it to bend in a curve-like shape. The top mold has incisions to give the 'elbows' air pockets, along with a channel and a circular part to allow for air pressure to travel throughout each leg smoothly. A thin mold in the shape of the legs was also created to allow for the mold to be closed, allowing for air pockets and channels.

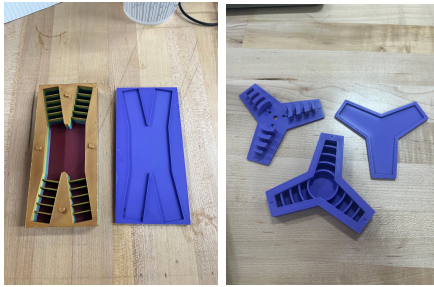


Figure 2: 3D printed molds

Once the CAD was made of the molds, they were then 3D printed using a Bambu Lab X1C Carbon printer, using PLA. Figure 2 shows the 3D printed body's molds on the left. A polymer was created by combining Dragon Skin 10 parts A and B, which was then poured into the molds and put into an oven to cure.



Figure 3: Final fabrication of the base

Once out of the oven, the silicone was carefully removed from the molds. The base of the body and the flat cap were glued together using Dragon Skin 10, and any holes or leaks in the body were patched with Dragon Skin 10 or Sil Pox. Four plastic tubings were inserted into the body, each one connected to a leg to provide the ability for air to be pressurized and therefore to actuate each leg. The two front legs' tubes were connected, and the two back legs' tubes were connected, then connected to syringes to

allow for simultaneous actuation to allow for the walking mechanism. The completed body is shown in Figure 3.

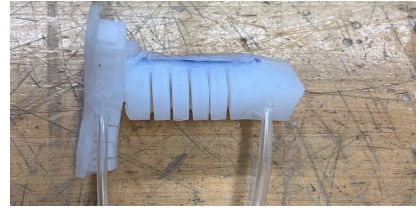


Figure 4: connection between body and grasper

Our grasping mechanism had two sub-parts: an arm for positioning our grabber and the grabber itself. The arm for positioning acted as a connection piece between the main body of the robot and the grasper. It was made identically to the body of the robot, however it is one leg chopped off the body, as shown in figure 4. A rubber band is attached to the two ends of this connection piece to add more strength to its function, which is to actuate from the vertical position curving down to allow the grasper to touch the ground below. The team tested different methods for a connection piece, including a TPE and fiberglass actuator, but decided on another pneu-net for more strength and durability. A plastic tube was inserted into the side of the connection piece to provide actuation.

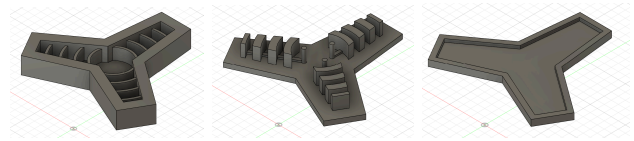


Figure 5: CAD of the grasper's molds

The grasper for the robot was designed and manufactured in the same way as the body of the robot. This design was chosen for the grasper because of silicone's strength and good functionality to squeeze onto objects. It was decided to use 3 arms in a circular pattern to allow the robot multi-directional access to objects and provide strength. Figure 5 shows the CAD models of the molds, and the right image of figure 2 shows the 3D printed molds for the grasper.

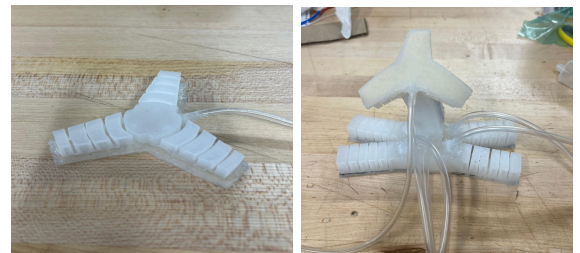


Figure 6: final grasper and final assembled robot

The left image in Figure 6 shows the final assembled grasper. All three arms of the grasper have

connecting air channels and only one tube inserted into the side of the silicone to allow for all 3 arms to actuate simultaneously and for the tubing to not interfere with this movement. The right image in Figure 6 shows the final assembled robot with all its components.

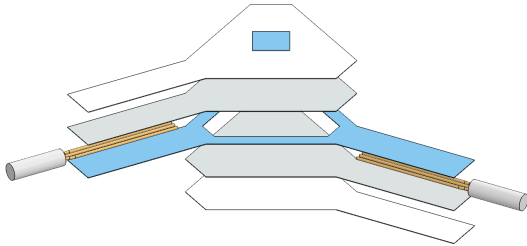


Figure 7: CAD of the Sensors Layers

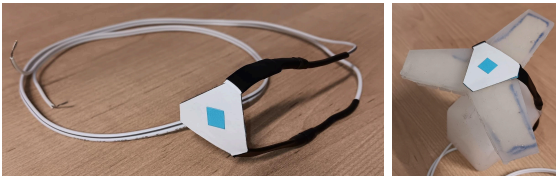


Figure 8: Fabricated Sensor and Mounted Location

Figure 7 shows the layer components of the sensor. The outer layers(white) of the sensor are made of printer paper adhered to aluminum foil(grey) using double sided tape. The middle layer(blue) is made of cardstock with a center cutout, and it is adhered to the outer layers using double sided tape. Finally, wires are held between the foil and middle layers using heat shrink and a square of cardstock is adhered to the contact surface of the sensor. Applying pressure to the sensor results in the two conductive layers touching and creating a closed surface, otherwise the sensor is an open circuit.

Originally, the outer layers were made using cardstock, but the sensor ended up being too stiff and insensitive for the smaller scale forces we were expecting. Additionally, the cardstock square was added because larger objects were resting on the outer frame of the middle layer and not closing the gap in the center. The square fixed this issue by making the center of the sensor make contact first, and close the circuit, before the outer edge. Finally, due to the materials used, the sensor is able to be bent but not elongated. In the context of this project, this limitation was deemed acceptable because the gripper palm was expected to have minimal bending and virtually no elongation. In other applications, the sensor could be fabricated using elastic materials to allow for elongation. However, a stiffening layer, which could affect elongation,

might be needed to prevent the outer layers from sagging and closing the circuit

The sensitivity of the sensor can be modified by using materials with different stiffnesses for the outer layers and changing the thickness of the middle layer. The combination of materials used for the final sensor, as seen in Figure 8, resulted in needing a minimum of 50g placed on the palm of the gripper to activate the sensor. Additionally, the final sensor was able to be bent approximately 90 degrees, while maintaining functionality. If the bend is sharper, such as a corner, then the sensor will perpetually trigger. However a sharp bend would be unexpected due to the material and design of the gripper.

V. MODELING COMPONENT

To model the movement of the pneu-net actuator, we used ABAQUS and the 3D cad files of the base actuator to simulate one leg under pressure. The setup for the modeling component was similar to that of Lab 5 with some slight differences. The geometry of the actuator was different as well as the number of parts used in the design. In the original lab, there were three separate pieces with two rectangles acting as the pneumatic base with fabric in between them. In our design, the fabric was embedded within the silicone layer meaning we could simplify the model. The material property values used in the modeling also are accurate to Dragon Skin 10. ABAQUS required the mass density, C10 and C20 values. The mass density was found on the product website at 1.07 g/cm^3 and the values for C10 and C20 were found using the Yeoh model on the Soft Robotics Material Database. C10 was 0.0547 and C20 was 0.0002. [3] In addition to the model and material changes, the pressure that was used to actuate the pneu-net leg was 0.025 MPa. As a result of the simulation, the deformation of a singular leg from the base is shown in Figure 9 below.

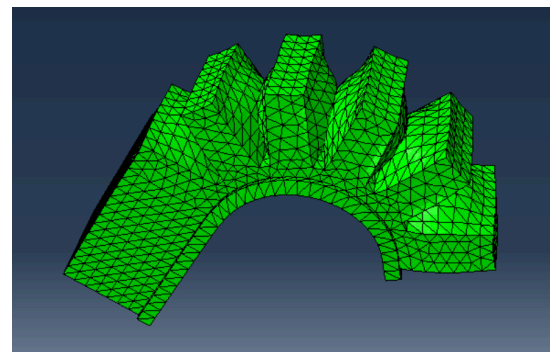


Figure 9: ABAQUS Modeling

VI. TESTING AND VALIDATION OF FUNCTIONALITY

To test and validate the functionality of our soft robot, we connected each pneumatic leg to a syringe that allowed us to pressurize and depressurize each leg in sequence. By doing this, the robot's movement was tested and a unique method of walking was developed. The movement was reminiscent of how a worm moves by lifting the back half of its body and then slowly raising the front half. By repeating this motion in sequence, the robot was able to slowly move forward using only pressure input. This validated the functionality of the base of the robots through its movement. To quantify the speed of the robot we measured how long it would take for it to move one full body and we got the result of 90 seconds. In addition to the speed of the robot, the angle of incline that it could scale was measured at 15 degrees.

To evaluate the strength of each pneumatic joint, we carried out a series of tests. For the leg component of the robot, we measured the maximum angle that the leg could reach while increasing the amount of weight. The maximum that an individual leg could hold was 300 grams. The results of this testing are summarized in Figure 10 below.

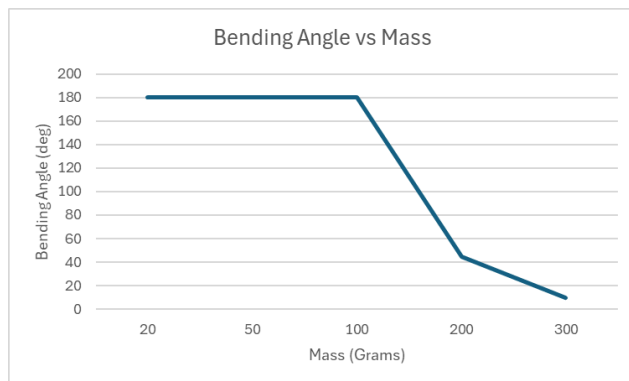


Figure 10: Bending Angle

Next, the gripper was evaluated by testing the amount of weight that it could grab and hold onto. In this experiment the shape of the weight being held was critical to the success of the strength of the gripper. The gripper design is more well suited for round objects that have as much surface contact with the palm as possible. Thus, to effectively evaluate the gripper, each weight was attached to a filled balloon that the gripper could more easily hold onto. Using this method, the maximum mass that could be held was 200 grams. The dependence on a reliably shaped object was a limitation on the functionality of the overall design of the gripper.

Finally, to test the strength of the arm of the robot, a weight was tied to a rope and was incrementally increased until the limit of its strength was reached. Due to the design of the pneu-net actuator, it is better suited for force actuation to reach down for an object and not for bringing it back up. To try to address this limitation in the design a rubber band was added to give supplemental strength. This resulted in a lower strength of the arm being able to only lift 70 grams.

VII. CONCLUSION AND BROADER IMPACT

Our robot represents a step forward in leveraging unique advantages of soft robot materials for planetary exploration. The soft actuation demonstrated both compliant locomotion and effective grasping capabilities. In testing, the robot proved resilient and functional, surviving a 20-foot drop without damage and navigating inclines of up to 15 degrees. The gripper held objects up to 200 grams, while its arm and legs demonstrated impressive bending angles and load capacity. Having a soft-robot working alongside a human (or astronaut) would improve safety of tasks requiring high compliance and low environmental impact, such as martian exploration with an element of human-robot collaboration. Whilst there are limitations, such as a dependency on object shape for effective grasping and the reduced lifting capacity, the challenges provide goals we can aim to achieve at future iterations. The broader impact of this project is in fields like space exploration. Soft robots such as Penu Robot offer safer, more adaptable solutions compared to traditional rigid robots, making them impactful for uncertain environments such as the moon, mars and beyond.

VIII. BIBLIOGRAPHY

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3. "Soft robotics materials database," Soft Robotics Materials Database, <https://soro-materials-database.onrender.com/> (accessed Dec. 10, 2024).

IX. APPENDIX

Link for Final Video:

https://drive.google.com/file/d/14JkFwwrXyf1XYTlvcH_MVWr77xyNkZkw/view?usp=sharing

